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Integral backstepping sliding mode control for quadrotor helicopter under external uncertain disturbances

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A B S T R A C T

In this paper, a novel nonlinear control strategy along with its simulation for a quadrotor helicopter is proposed. The normal 6-DOF dynamic model of the quadrotor based on the Newton–Euler formula as well as the model with external uncertain disturbances are established. Considering the underactuated and strongly coupled characteristics of quadrotor helicopter, we design a nonlinear control method by using integral backstepping combined with the sliding mode control (integral BS-SMC) to stabilize the quadrotor attitude and to accomplish the task of trajectory tracking. The designed controllers based on the hierarchical control scheme can be divided into rotational controller and translational controller and their stability are validated by the Lyapunov stability theorem. By means of the proposed controllers, the chattering phenomenon and discontinuousness of control inputs faced by traditional sliding mode control (SMC) can be avoided. The feasibility of the control approach presented in this paper is verified by the simulations under different scenarios. The results show that the nonlinear control method not only has a better tracking performance than others but also has a higher robustness when unknown disturbances occur.

1. Introduction

Unmanned aerial vehicles (UAVs), especially quadrotor helicopters, have been developed for recent decades and have attracted much research interest in the field of commercial and civil missions due to its security for human and flexibility for application. Compared with traditional manned airplanes and the unmanned fixed-wing flight vehicles, the quadrortor helicopters have many significant advantages, e.g., vertical take-off and landing (VTOL), stable hovering, convenient portability, small size, low cost, and versatile feature. Therefore, quadrotor helicopters are eligible to execute various of complex tasks like military surveillance, agricultural investigation, disaster assessment and commercial photography [1–3]. To fly autonomously and to meet the task requirements with high reliability, attitude controlling and path following are among the most important technologies. However, due to the underactuated characteristics of quadrotor in nature as well as its high nonlinearity with strongly coupled dynamics, the control of quadrotor becomes more difficult than other flight vehicles. Meanwhile, external uncertain disturbances also bring a challenge for controlling the quadrotor helicopters. Thus, this paper focuses on the quadrotor modeling and the controller designing in order to achieve higher tracking accuracy and better robustness against external disturbances.

For the purpose of simpleness and practicability, studies adopted linear control methods such as proportional-integral-derivative (PID) and linear quadratic regulator (LQR) [4–8] in designing the quadrotor controller in early quadrotor helicopters research stage. The linear control technology was used to stabilize the attitude of the quadrotor by linearizing the dynamic model and neglecting the unimportant factors. For this reason, the performances of the quadrotor, such as the tracking accuracy, the robustness against the disturbances, etc., were poor and even not acceptable.

With the development of control technology and the wide use of quadrotor, an increasing number of nonlinear control methods have been applied and have carried out better effectiveness in some aspects.

The sliding mode control (SMC) is a main type of variable structure control which turns out to be with high simplicity and robustness against disturbances. The controller in SMC switches between two structures to make the system states to a pre-defined manifold which is defined in the state space or the error state space of the system. Authors of [9] divided the quadrotor system into a fully-actuated subsystem and an underactuated subsystem, meanwhile, a sliding mode control method was proposed to stabilize the systems and to make the quadrotor reach a desired position

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with a desired yaw angle while keeping the pitch and the roll angles zero. In [10,11], sliding mode control approach was presented to stabilize a series of underactuated systems in cascaded model. The designed controller was insensitive to the model errors and uncertainties and was able to stabilize the system globally. A kind of robust sliding mode controller was applied in [12] for a highly nonlinear UAV. Three sliding mode controllers were designed for three second order subsystems decoupled from the nonlinear system. However, the unacceptable phenomena in SMC are the chattering and the discontinuousness of control inputs. On the other hand, in order to estimate and compensate the effects of the unknown external disturbances, [13–15] presented classical and higher order sliding mode control driven by sliding mode disturbance observer (SMC-SMDO) approaches. [16] summarized the previous works on SMC-SMDO and designed a robust position and attitude controller only based on the knowledge of the boundaries of the disturbances. An adaptive fuzzy integral sliding mode controller was proposed in [17] aiming at reducing the effect of the norm bounded and external disturbances. The validity of this method was proved by using an uncertain mechanical system. In addition, only the matched unknown disturbances can be compensated by the sliding mode controller.

The backstepping (BS) control methodology has been widely used in nonlinear systems under matched and mismatched disturbances in recent years. The main idea in backstepping technology is to choose appropriate functions of state variables as virtual control inputs recursively. Once the final control input is determined, the stability of the whole system is ensured. The attitude system of quadrotor was stabilized by a backstepping controller in [18, 19]. [20] focused on the attitude control problem using a hybrid backstepping controller based on the Frenet–Serret theory. The results showed that the presented controller provided a robust flight when facing the wind disturbances. [21] designed an adaptive controller combined with backstepping and parameters adaptive control for trajectory tracking problem. Unfortunately, conventional backstepping can only deal with the constant or slowly changing uncertainties. Considering the characteristic of the sliding mode and backstepping, it is reasonable to utilize the advantages of both control methods and to avoid the shortcomings mentioned above. However, researches in this field applied to quadrotor helicopter are quite few. An adaptive sliding mode backstepping control was designed in [22] for the wheeled mobile manipulator. The results showed that the parametric uncertainties can be compensated and the bounded disturbances were well suppressed. In [23], the attitude controller was developed in the backstepping method and the adaptive sliding mode is presented in the final step to deal with the inertia matrix uncertainty and time-varying external disturbances. An integral backstepping sliding mode controller is proposed for swing-up and stabilization of a Cart–Pendulum system in [24]. The simulation results showed that the controller was able to reject both matched and mismatched disturbances.

In this paper, a novel control methodology combined with integral backstepping and sliding mode is proposed for the quadrotor trajectory tracking problem under external disturbances. The structure of this paper is organized as follows: the classical and disturbed 6-DOF dynamic model of quadrotor helicopter as well as the model of actuators are established in Section 2. In Section 3, the nonlinear control strategy is applied to rotational subsystem and translational subsystem respectively based on the hierarchical control scheme according to the dynamic characteristics. Meanwhile, the stabilization of the nonlinear control method presented in this paper is validated by the candidate Lyapunov function. In Section 4, two numerical simulation scenarios are carried out to demonstrate the validity of the control method and to illustrate the performance of path following compared with other control methods. Major conclusions are drawn in Section 5.

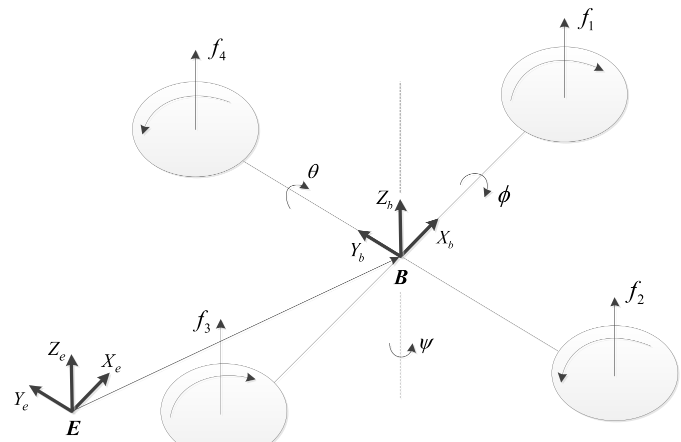

Fig. 1. Quadrotor helicopter scheme.

2. The model of quadrotor system

The typical quadrotor consists of a rigid body equipped with four rotors as shown in Fig. 1. The two pairs of rotors $(1, 3)$ and $(2, 4)$ turn in opposite direction to balance or produce the yaw motion. The vertical motion is achieved by increasing or decreasing the speed of rotors together. The forward (or lateral) motion corresponding to the pitch (or roll) motion is created by changing the 1 and 3 (or 2 and 4) propeller's speed conversely. The classical and disturbed 6-DOF dynamic model of quadrotor helicopter as well as the model of actuators are presented in this section.

In order to establish the dynamic model of quadrotor in general, some assumptions are introduced as follows:

- The quadrotor is a rigid body with an invariant mass.
- The quadrotor is symmetric.
- The lifts and reactive moment produced by four rotors respectively are proportional to the square of their rotational speed.

2.1. Classical quadrotor dynamic model

The orthogonal right-handed reference frames are defined as follows and shown in Fig. 1.

- $\boldsymbol{E} = (X_e, Y_e, Z_e)^{\mathrm{T}}$ defines the inertial frame.
- $\boldsymbol{B} = (X_b, Y_b, Z_b)^{\mathrm{T}}$ defines the body-fixed frame.

Supposed that the vectors $\boldsymbol{\xi} = [x, y, z]^{\mathrm{T}}$ (frame $\boldsymbol{E}$) and $\boldsymbol{\eta} = [\phi, \theta, \psi]^{\mathrm{T}}$ denote the positions and the attitude angles (i.e., roll angle, pitch angle and yaw angle) respectively of the quadrotor helicopter. $\boldsymbol{R}_E^B \in SO(3)$ is the orthogonal rotation matrix from the inertial frame to the body-fixed frame.

$$\boldsymbol{R}_E^B = \begin{bmatrix} \cos\psi\cos\theta & \cos\theta\sin\psi & -\sin\theta \\ \cos\psi\sin\phi\sin\theta - \cos\phi\sin\psi & \cos\phi\cos\psi + \sin\phi\sin\psi\sin\theta & \cos\theta\sin\phi \\ \sin\phi\sin\psi + \cos\phi\cos\psi\sin\theta & \cos\phi\sin\psi\sin\theta - \cos\psi\sin\phi & \cos\phi\cos\theta \end{bmatrix} \tag{1}$$

The equations which describe the dynamics of quadrotor helicopter and involve three translational speed $\boldsymbol{V} = [u, v, w]^{\mathrm{T}}$ (frame $\boldsymbol{E}$) and three angular rate $\boldsymbol{\Omega} = [p, q, r]^{\mathrm{T}}$ (frame $\boldsymbol{B}$) are based on the Newton–Euler formula with 6-DOF, written as:

$$\dot{\boldsymbol{\xi}} = \boldsymbol{V} \tag{2}$$

$$m\dot{\boldsymbol{V}} = \boldsymbol{R}_B^E \boldsymbol{T}^B + \boldsymbol{G}^E \tag{3}$$

$$\dot{\boldsymbol{\eta}} = \boldsymbol{W}\boldsymbol{\Omega} \tag{4}$$

$$\boldsymbol{J}\dot{\boldsymbol{\Omega}} = -\boldsymbol{\Omega} \times \boldsymbol{J}\boldsymbol{\Omega} + \boldsymbol{M}^B \tag{5}$$

where:

$$\boldsymbol{W}=\begin{bmatrix}1 & \sin\phi\tan\theta & \cos\phi\tan\theta\\ 0 & \cos\phi & -\sin\phi\\ 0 & \sin\phi/\cos\theta & \cos\phi/\cos\theta\end{bmatrix}$$

And $\boldsymbol{R}_B^E=(\boldsymbol{R}_E^B)^{\mathrm{T}}$, $\boldsymbol{J}=\mathrm{diag}(J_x, J_y, J_z)$ denotes the moment of inertial of quadrotor helicopter, $\boldsymbol{G}^E=(0,0,-mg)^{\mathrm{T}}$ is the gravity, $\boldsymbol{T}^B=(0,0,L)^{\mathrm{T}}$ is the total lift produced by four rotors:

$$L=\sum_{i=1}^{4} f_i=b\sum_{i=1}^{4}\omega_i^2 \quad (i=1,2,3,4) \tag{6}$$

where ω_i is the speed of rotor i respectively and b is a positive constant called thrust factor.

The control torque $\boldsymbol{M}^B$ developed by four rotors according to the body-fixed frame is:

$$\boldsymbol{M}^B=\begin{bmatrix}M_x^B\\ M_y^B\\ M_z^B\end{bmatrix}=\begin{bmatrix}d(f_4-f_2)\\ d(f_3-f_1)\\ k_d(\omega_1^2+\omega_3^2-\omega_2^2-\omega_4^2)\end{bmatrix} \tag{7}$$

where d is the distance between the quadrotor center of mass and the axis of the propeller and k_d is the drag factor.

To demonstrate the relationship between the speed of the rotors and the control inputs, we put the Eq. (6) and Eq. (7) together as follows:

$$\begin{bmatrix}L\\ M_x^B\\ M_y^B\\ M_z^B\end{bmatrix}=\begin{bmatrix}b & b & b & b\\ 0 & -db & 0 & db\\ -db & 0 & db & 0\\ k_d & -k_d & k_d & -k_d\end{bmatrix}\begin{bmatrix}\omega_1^2\\ \omega_2^2\\ \omega_3^2\\ \omega_4^2\end{bmatrix} \tag{8}$$

Consequently, we get the complete quadrotor dynamics written as two parts:

$$\begin{cases}\dot{u}=\dfrac{(\sin\phi\sin\psi+\cos\phi\cos\psi\sin\theta)L}{m}\\ \dot{v}=\dfrac{(\cos\phi\sin\psi\sin\theta-\cos\psi\sin\phi)L}{m}\\ \dot{w}=\dfrac{(\cos\phi\cos\theta)L-mg}{m}\end{cases} \tag{9}$$

$$\begin{cases}\dot{p}=\dot{\theta}\dot{\psi}\left(\dfrac{J_y-J_z}{J_x}\right)+\dfrac{M_x^B}{J_x}\\ \dot{q}=\dot{\phi}\dot{\psi}\left(\dfrac{J_z-J_x}{J_y}\right)+\dfrac{M_y^B}{J_y}\\ \dot{r}=\dot{\phi}\dot{\theta}\left(\dfrac{J_x-J_y}{J_z}\right)+\dfrac{M_z^B}{J_z}\end{cases} \tag{10}$$

where Eq. (9) and Eq. (10) represent translational dynamics and rotational dynamics of the quadrotor respectively.

2.2. Disturbed quadrotor dynamic model

The disturbed vectorial dynamic model of the quadrotor combining the classical model shown in Eq. (3) and Eq. (5) and the external uncertain disturbances $\boldsymbol{d}=[\boldsymbol{d}_1,\boldsymbol{d}_2]^{\mathrm{T}}$ is written as:

$$m\dot{\boldsymbol{V}}=\boldsymbol{R}_B^E\boldsymbol{T}^B+\boldsymbol{G}^E+\boldsymbol{d}_1 \tag{11}$$

$$\boldsymbol{J}\dot{\boldsymbol{\Omega}}=\boldsymbol{M}^B-\boldsymbol{\Omega}\times\boldsymbol{J}\boldsymbol{\Omega}+\boldsymbol{d}_2 \tag{12}$$

where $\boldsymbol{d}_1$ is modeled as the force disturbances and $\boldsymbol{d}_2$ means the moment disturbances.

The external disturbances can be characterized by two parts:

$$\boldsymbol{d}_i=\boldsymbol{d}_{i,cont}+\boldsymbol{d}_{i,sto} \quad (i=1,2) \tag{13}$$

- $\boldsymbol{d}_{i,cont}$ (e.g., static wind, unmodeled dynamic errors) represents the unknown constant disturbances changing slowly along time.
- $\boldsymbol{d}_{i,sto}$ (e.g., stochastic wind, uncertain measurement noise) represents the fast varying and unknown stochastic disturbances with the assumption that corresponds to uniform distribution.

Remark 1. The disturbances $\boldsymbol{d}_1$ and $\boldsymbol{d}_2$ are unknown but norm bounded such that $\|\boldsymbol{d}_i\|\le B_i$.

2.3. Actuators dynamic model

Given that actuator dynamics is not the main emphasis of our research, we describe the dynamics of each actuator approximately by the following first-order inertia element:

$$\frac{\omega(s)}{\omega_r(s)}=\frac{1}{T_{act}s+1} \tag{14}$$

where T_{act} is the time constant of the actuator dynamics, ω_r and ω are desired rotation speed and actual rotation speed of the actuator.

Remark 2. As for the stochastic characteristic of the actuator actual rotation speed, the effectiveness of this factor can be cast within the framework of the Eqs. (11)–(12). Moreover, we assume that $0\le\omega_r\le\omega_{\max}$, where $\omega_{\max}$ is the maximum rotation speed of the practical electric motor.

Remark 3. Considering the different time scale, the control laws of the actuator servo system and the quadrotor system are designed separately. Thus, we neglect the controller for the actuator system in this paper and such first-order inertia element assumption is reasonable.

3. Nonlinear controller design and stability analysis

In this section, a novel nonlinear control strategy is addressed with the goal of tracking the reference path under the constant and unknown stochastic disturbances. In order to exploit the quadrotor characteristics that the dynamic equations contain two components, translation dynamics and rotation dynamics, we adopt hierarchical control scheme as shown in Fig. 2. The hierarchical control scheme is used to illustrate the main process of trajectory tracking. In this hierarchical control scheme, the rotational subsystem represented by the inner-loop is used to stabilize the attitudes of the quadrotor. Meanwhile, the position of quadrotor is controlled by the outer-loop which means the translational subsystem.

Compared with the traditional backstepping control, the integral backstepping control combined with the sliding mode control strategy presented in this paper increases the robustness against external disturbances, avoiding the discontinuous control signal and improving the tracking accuracy.

3.1. Rotational subsystem controller

The rotational subsystem, i.e., inner-loop in control scheme is used to stabilize the attitude of the quadrotor helicopter.

First, consider the virtual system:

$$\dot{\boldsymbol{\eta}}=\boldsymbol{\alpha}_1 \tag{15}$$

where $\boldsymbol{\alpha}_1$ is the virtual control input.

Define the angle tracking error as:

$$\boldsymbol{e}_1=\boldsymbol{\eta}_d-\boldsymbol{\eta}+\boldsymbol{K}_1\int(\boldsymbol{\eta}_d-\boldsymbol{\eta})\mathrm{d}t \tag{16}$$

$\boldsymbol{K}_1\in\mathbf{R}^{3\times3}$ is a positive definite diagonal integral coefficient matrix.

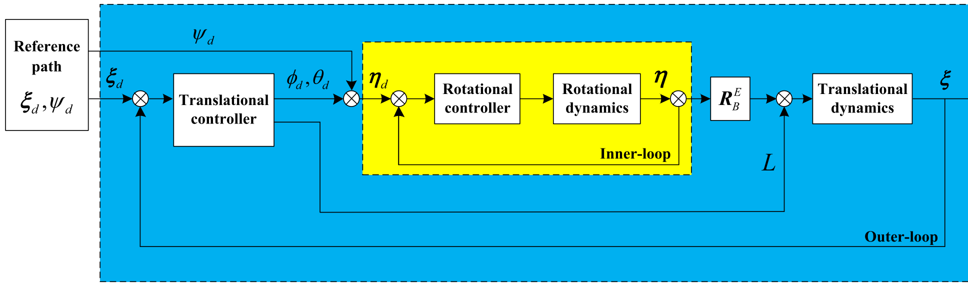

Fig. 2. Hierarchical control scheme.

Considering the candidate Lyapunov function:

$$V_1(t) = \frac{1}{2}\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1 \tag{17}$$

The time derivative is given by:

$$\begin{aligned}\dot{V}_1(t) &= \boldsymbol{e}_1^{\mathrm{T}}\dot{\boldsymbol{e}}_1 \\ &= \boldsymbol{e}_1^{\mathrm{T}}\big(\dot{\boldsymbol{\eta}}_d - \dot{\boldsymbol{\eta}} + \boldsymbol{K}_1(\boldsymbol{\eta}_d - \boldsymbol{\eta})\big) \\ &= \boldsymbol{e}_1^{\mathrm{T}}\big(\dot{\boldsymbol{\eta}}_d - \boldsymbol{\alpha}_1 + \boldsymbol{K}_1(\boldsymbol{\eta}_d - \boldsymbol{\eta})\big)\end{aligned} \tag{18}$$

To make the virtual system stabilized, the virtual control input $\boldsymbol{\alpha}_1$ should be:

$$\boldsymbol{\alpha}_1 = \dot{\boldsymbol{\eta}}_d + \boldsymbol{A}_1\boldsymbol{e}_1 + \boldsymbol{K}_1(\boldsymbol{\eta}_d - \boldsymbol{\eta}) \tag{19}$$

where $\boldsymbol{A}_1 \in \mathbf{R}^{3\times 3}$ is a positive definite diagonal matrix.

Thus, we can get:

$$\dot{V}_1(t) = -\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{A}_1\boldsymbol{e}_1 < 0, \quad \forall \boldsymbol{e}_1 \neq 0 \tag{20}$$

then, $\boldsymbol{e}_1 = 0$ is globally asymptotically stable.

From Eq. (12), we introduce a new virtual system as follows:

$$\dot{\boldsymbol{\Omega}} = \boldsymbol{f}(\boldsymbol{\Omega}) + \boldsymbol{J}^{-1}\boldsymbol{\alpha}_2 + \boldsymbol{J}^{-1}\boldsymbol{d}_2 \tag{21}$$

where the vectorial function $\boldsymbol{f}(\boldsymbol{\Omega})$ is the first part of Eq. (5) and the virtual control input $\boldsymbol{\alpha}_2 \in \mathbf{R}^3$ equals to the control torque $\boldsymbol{M}^B$.

Thus, we can get the sliding surface as:

$$\begin{aligned}\boldsymbol{e}_2 &= \boldsymbol{\alpha}_1 - \dot{\boldsymbol{\eta}} \\ &= \dot{\boldsymbol{\eta}}_d + \boldsymbol{A}_1\boldsymbol{e}_1 + \boldsymbol{K}_1(\boldsymbol{\eta}_d - \boldsymbol{\eta}) - \dot{\boldsymbol{\eta}} \\ &= \dot{\boldsymbol{e}}_1 + \boldsymbol{A}_1\boldsymbol{e}_1 \\ &= \boldsymbol{s}_1\end{aligned} \tag{22}$$

The candidate Lyapunov function is:

$$V_2(t) = \frac{1}{2}\big(\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1 + \boldsymbol{s}_1^{\mathrm{T}}\boldsymbol{s}_1\big) \tag{23}$$

Its time derivative is:

$$\begin{aligned}\dot{V}_2(t) &= \boldsymbol{e}_1^{\mathrm{T}}\dot{\boldsymbol{e}}_1 + \boldsymbol{s}_1^{\mathrm{T}}\dot{\boldsymbol{s}}_1 \\ &= \boldsymbol{e}_1^{\mathrm{T}}(-\boldsymbol{A}_1\boldsymbol{e}_1 + \boldsymbol{s}_1) + \boldsymbol{s}_1^{\mathrm{T}}(\dot{\boldsymbol{\alpha}}_1 - \ddot{\boldsymbol{\eta}}) \\ &= -\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{A}_1\boldsymbol{e}_1 + \boldsymbol{s}_1^{\mathrm{T}}\big(\ddot{\boldsymbol{\eta}}_d + \boldsymbol{A}_1(-\boldsymbol{A}_1\boldsymbol{e}_1 + \boldsymbol{s}_1) \\ &\quad + \boldsymbol{K}_1(\dot{\boldsymbol{\eta}}_d - \dot{\boldsymbol{\eta}}) - \ddot{\boldsymbol{\eta}} + \boldsymbol{e}_1\big)\end{aligned} \tag{24}$$

To stabilize the virtual system, we substitute Eq. (21) into Eq. (24) and obtain the virtual control input as well as the control torque as follows:

$$\begin{aligned}\boldsymbol{\alpha}_2 = \boldsymbol{J}\big[&\boldsymbol{A}_1(-\boldsymbol{A}_1\boldsymbol{e}_1 + \boldsymbol{s}_1) + \boldsymbol{K}_1(\dot{\boldsymbol{\eta}}_d - \boldsymbol{\alpha}_1 + \boldsymbol{s}_1) - \boldsymbol{f}(\boldsymbol{\Omega}) + \boldsymbol{e}_1 \\ &+ \varepsilon_1\mathrm{sign}(\boldsymbol{s}_1) + \boldsymbol{Q}_1\boldsymbol{s}_1\big]\end{aligned} \tag{25}$$

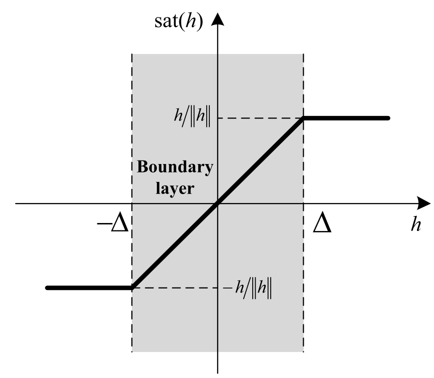

Fig. 3. Saturation function.

where $\varepsilon_1 \in \mathbf{R}^+$ and $\boldsymbol{Q}_1 \in \mathbf{R}^{3\times 3}$ are positive real number and positive definite diagonal matrix. And the signum function is defined as:

$$\mathrm{sign}(h) = \begin{cases} h/\|h\| & h > 0 \\ 0 & h = 0 \\ -h/\|h\| & h < 0 \end{cases} \tag{26}$$

In order to reduce the chatting characteristic in the sliding mode control, boundary layer method is used to replace the signum function in this paper. The saturation function is defined in Eq. (27) and illustrated in Fig. 3.

$$\mathrm{sat}(h) = \begin{cases} h/\|h\| & h > \Delta \\ 0 & h = 0 \\ -h/\|h\| & h < \Delta \\ (h/\Delta\|h\|)h & \text{otherwise} \end{cases} \tag{27}$$

where Δ is a positive constant.

Thus, we can rewrite the control law as:

$$\begin{aligned}\boldsymbol{\alpha}_2 = \boldsymbol{J}\big[&\boldsymbol{A}_1(-\boldsymbol{A}_1\boldsymbol{e}_1 + \boldsymbol{s}_1) + \boldsymbol{K}_1(\dot{\boldsymbol{\eta}}_d - \boldsymbol{\alpha}_1 + \boldsymbol{s}_1) - \boldsymbol{f}(\boldsymbol{\Omega}) + \boldsymbol{e}_1 \\ &+ \varepsilon_1\mathrm{sat}(\boldsymbol{s}_1) + \boldsymbol{Q}_1\boldsymbol{s}_1\big]\end{aligned} \tag{28}$$

Theorem 1. *Given the disturbed dynamic model described in Eq. (12), together with the Remark 1, the control law in Eq. (28) can globally asymptotically stabilize the rotational subsystem in the presence that:*

$$\varepsilon_1 - \big\|\boldsymbol{J}^{-1}\boldsymbol{d}_2\big\| \geq 0 \tag{29}$$

Proof. Combining Eq. (24) and Eq. (28), we can obtain the time derivative of V_2:

$$\begin{aligned}\dot{V}_2(t) &= -\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{A}_1\boldsymbol{e}_1 - \boldsymbol{s}_1^{\mathrm{T}}\boldsymbol{J}^{-1}\boldsymbol{d}_2 - \boldsymbol{s}_1^{\mathrm{T}}\varepsilon_1\mathrm{sat}(\boldsymbol{s}_1) - \boldsymbol{s}_1^{\mathrm{T}}\boldsymbol{Q}_1\boldsymbol{s}_1 \\ &\leq -\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{A}_1\boldsymbol{e}_1 - \boldsymbol{s}_1^{\mathrm{T}}\boldsymbol{Q}_1\boldsymbol{s}_1 + \big\|\boldsymbol{J}^{-1}\boldsymbol{d}_2\big\|\|\boldsymbol{s}_1\| - \varepsilon_1\|\boldsymbol{s}_1\| \\ &= -\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{A}_1\boldsymbol{e}_1 - \boldsymbol{s}_1^{\mathrm{T}}\boldsymbol{Q}_1\boldsymbol{s}_1 - \big(\varepsilon_1 - \big\|\boldsymbol{J}^{-1}\boldsymbol{d}_2\big\|\big)\|\boldsymbol{s}_1\| \\ &< 0, \quad \forall \boldsymbol{e}_1 \neq 0,\ \boldsymbol{s}_1 \neq 0\end{aligned} \tag{30}$$

Thus, $\boldsymbol{e}_2 = 0$ is globally asymptotically stable. □

3.2. Translational subsystem controller

The translational subsystem is the outer-loop in the control scheme. The main goal of the translational subsystem controller is to get the desired commands of pitch angle, roll angle and lift produced by four rotors in order to track the reference path.

Consider the virtual system:

$$\dot{\boldsymbol{\xi}} = \boldsymbol{\alpha}_3 \tag{31}$$

The position tracking error is defined as:

$$\boldsymbol{e}_3 = \boldsymbol{\xi}_d - \boldsymbol{\xi} + \boldsymbol{K}_2 \int (\boldsymbol{\xi}_d - \boldsymbol{\xi})\mathrm{d}t \tag{32}$$

where $\boldsymbol{K}_2 \in \mathbf{R}^{3\times 3}$ is a positive definite diagonal integral coefficient matrix.

Then, the candidate Lyapunov function is:

$$V_3(t) = \frac{1}{2}\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3 \tag{33}$$

The time derivative can be obtained as:

$$\begin{aligned}\dot{V}_3(t) &= \boldsymbol{e}_3^{\mathrm{T}}\dot{\boldsymbol{e}}_3 \\ &= \boldsymbol{e}_3^{\mathrm{T}}\big(\dot{\boldsymbol{\xi}}_d - \dot{\boldsymbol{\xi}} + \boldsymbol{K}_2(\boldsymbol{\xi}_d - \boldsymbol{\xi})\big) \\ &= \boldsymbol{e}_3^{\mathrm{T}}\big(\dot{\boldsymbol{\xi}}_d - \boldsymbol{\alpha}_3 + \boldsymbol{K}_2(\boldsymbol{\xi}_d - \boldsymbol{\xi})\big)\end{aligned} \tag{34}$$

The system can be stabilized when the virtual control $\boldsymbol{\alpha}_3$ is:

$$\boldsymbol{\alpha}_3 = \dot{\boldsymbol{\xi}}_d + \boldsymbol{A}_2\boldsymbol{e}_3 + \boldsymbol{K}_2(\boldsymbol{\xi}_d - \boldsymbol{\xi}) \tag{35}$$

$\boldsymbol{A}_2 \in \mathbf{R}^{3\times 3}$ is a positive definite diagonal matrix.

Substitute Eq. (35) into Eq. (34), we yield:

$$\dot{V}_3(t) = -\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{A}_2\boldsymbol{e}_3 < 0, \quad \forall \boldsymbol{e}_3 \neq 0 \tag{36}$$

thus, $\boldsymbol{e}_3 = 0$ is globally asymptotically stable.

From the translational dynamic written as in Eq. (11), another virtual system is:

$$\dot{\boldsymbol{V}} = \boldsymbol{\alpha}_4 + \frac{1}{m}\boldsymbol{d}_1 \tag{37}$$

where $\boldsymbol{\alpha}_4 = [U_1, U_2, U_3]^{\mathrm{T}} \in \mathbf{R}^3$ is the virtual control input.

Defining the sliding surface:

$$\begin{aligned}\boldsymbol{e}_4 &= \boldsymbol{\alpha}_3 - \dot{\boldsymbol{\xi}} \\ &= \dot{\boldsymbol{\xi}}_d + \boldsymbol{A}_2\boldsymbol{e}_3 + \boldsymbol{K}_2(\boldsymbol{\xi}_d - \boldsymbol{\xi}) - \dot{\boldsymbol{\xi}} \\ &= \dot{\boldsymbol{e}}_3 + \boldsymbol{A}_2\boldsymbol{e}_3 \\ &= \boldsymbol{s}_2\end{aligned} \tag{38}$$

Based on the position tracking error and the sliding surface, we can get the candidate Lyapunov function:

$$V_4(t) = \frac{1}{2}\big(\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3 + \boldsymbol{s}_2^{\mathrm{T}}\boldsymbol{s}_2\big) \tag{39}$$

Thus, its time derivative can be obtained:

$$\begin{aligned}\dot{V}_4(t) &= \boldsymbol{e}_3^{\mathrm{T}}\dot{\boldsymbol{e}}_3 + \boldsymbol{s}_2^{\mathrm{T}}\dot{\boldsymbol{s}}_2 \\ &= \boldsymbol{e}_3^{\mathrm{T}}(-\boldsymbol{A}_2\boldsymbol{e}_3 + \boldsymbol{s}_2) + \boldsymbol{s}_2^{\mathrm{T}}(\dot{\boldsymbol{\alpha}}_3 - \ddot{\boldsymbol{\xi}}) \\ &= -\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{A}_2\boldsymbol{e}_3 + \boldsymbol{s}_2^{\mathrm{T}}\Big(\ddot{\boldsymbol{\xi}}_d + \boldsymbol{A}_2(-\boldsymbol{A}_2\boldsymbol{e}_3 + \boldsymbol{s}_2) \\ &\quad + \boldsymbol{K}_2(\dot{\boldsymbol{\xi}}_d - \dot{\boldsymbol{\xi}}) - \boldsymbol{\alpha}_4 - \frac{1}{m}\boldsymbol{d}_1 + \boldsymbol{e}_3\Big)\end{aligned} \tag{40}$$

In order to stabilize the system, the virtual control input as well as using the saturation function should be as follows:

$$\begin{aligned}\boldsymbol{\alpha}_4 &= \boldsymbol{A}_2(-\boldsymbol{A}_2\boldsymbol{e}_3 + \boldsymbol{s}_2) + \boldsymbol{K}_2(\dot{\boldsymbol{\xi}}_d - \boldsymbol{\alpha}_3 + \boldsymbol{s}_2) + \boldsymbol{e}_3 \\ &\quad + \varepsilon_2 \mathrm{sat}(\boldsymbol{s}_2) + \boldsymbol{Q}_2\boldsymbol{s}_2\end{aligned} \tag{41}$$

where $\varepsilon_2 \in \mathbf{R}^+$ and $\boldsymbol{Q}_2 \in \mathbf{R}^{3\times 3}$ are positive real number and positive definite diagonal matrix.

Theorem 2. *Given the disturbed dynamic model described in Eq. (11), together with the Remark 1, the control law in Eq. (41) can globally asymptotically stabilize the translational subsystem in the presence that:*

$$\varepsilon_2 - \frac{1}{m}\|\boldsymbol{d}_1\| \geq 0 \tag{42}$$

Proof. Combining Eq. (40) and Eq. (41), we can rewrite the time derivative of V_4:

$$\begin{aligned}\dot{V}_4(t) &= -\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{A}_2\boldsymbol{e}_3 - \frac{1}{m}\boldsymbol{s}_2^{\mathrm{T}}\boldsymbol{d}_1 - \boldsymbol{s}_2^{\mathrm{T}}\varepsilon_2\mathrm{sat}(\boldsymbol{s}_2) - \boldsymbol{s}_2^{\mathrm{T}}\boldsymbol{Q}_2\boldsymbol{s}_2 \\ &\leq -\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{A}_2\boldsymbol{e}_3 - \boldsymbol{s}_2^{\mathrm{T}}\boldsymbol{Q}_2\boldsymbol{s}_2 + \frac{1}{m}\|\boldsymbol{d}_1\|\|\boldsymbol{s}_2\| - \varepsilon_2\|\boldsymbol{s}_2\| \\ &= -\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{A}_2\boldsymbol{e}_3 - \boldsymbol{s}_2^{\mathrm{T}}\boldsymbol{Q}_2\boldsymbol{s}_2 - \left(\varepsilon_2 - \frac{1}{m}\|\boldsymbol{d}_1\|\right)\|\boldsymbol{s}_2\| \\ &< 0, \quad \forall \boldsymbol{e}_3 \neq 0,\ \boldsymbol{s}_2 \neq 0\end{aligned} \tag{43}$$

Thus, $\boldsymbol{e}_4 = 0$ is globally asymptotically stable. □

So far, we get the virtual control inputs for the translational subsystem. Then, the commands of pitch angle, roll angle and lift will be designed in functions of virtual control inputs.

Combining Eq. (3) and virtual control input $\boldsymbol{\alpha}_4$, we yield:

$$\begin{bmatrix}\dot{u}\\ \dot{v}\\ \dot{w}\end{bmatrix} = \boldsymbol{R}_B^E\begin{bmatrix}0\\ 0\\ L/m\end{bmatrix} + \begin{bmatrix}0\\ 0\\ -g\end{bmatrix} = \begin{bmatrix}U_1\\ U_2\\ U_3\end{bmatrix} \tag{44}$$

After some simple algebraic transformations, we can get the expressions for desired angles as follows:

$$\begin{cases}\phi_d = \sin^{-1}\left(\dfrac{U_1\sin\psi_d - U_2\cos\psi_d}{\sqrt{U_1^2 + U_2^2 + (U_3+g)^2}}\right) \\ \theta_d = \tan^{-1}\left(\dfrac{U_1\cos\psi_d + U_2\sin\psi_d}{U_3 + g}\right)\end{cases} \tag{45}$$

where ψ_d is the desired yaw angle given by the reference path.

And the lift generated by four rotors can be computed as:

$$L = \sqrt{m^2\big(U_1^2 + U_2^2 + (U_3+g)^2\big)} \tag{46}$$

3.3. The closed-loop quadrotor system stability analysis

The stability of the closed-loop quadrotor system is declared by the following theorem:

Theorem 3. *According to the hierarchical control scheme shown in Fig. 2, together with the control laws presented in Eq. (28) and Eq. (41), asymptotic translational tracking as well as rotational tracking are achieved in the sense that:*

$$\lim_{t\to 0} \boldsymbol{e}_1(t), \boldsymbol{e}_2(t), \boldsymbol{e}_3(t), \boldsymbol{e}_4(t) = 0 \tag{47}$$

where $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, $\boldsymbol{e}_3$ and $\boldsymbol{e}_4$ are defined as above.

Table 1
Physical parameters of the quadrotor helicopter.

Symbol	Description	Value (unit)
m	Mass of quadrotor	0.25 (kg)
d	Distance between mass center and propeller axis	0.20 (m)
J_x	Inertial moment along x-axis	2.35×10^{-3} (kg·m^2)
J_y	Inertial moment along y-axis	2.35×10^{-3} (kg·m^2)
J_z	Inertial moment along z-axis	5.26×10^{-2} (kg·m^2)
b	Thrust factor	6.13×10^{-5} (N·s^2)
k_d	Drag factor	2.5×10^{-6} (N·m·s^2)

Table 2
Reference flight trajectory parameters.

Time (s)	Initial position (m)	Terminal position (m)
$0 \sim 10$	[0, 0, 0]	[5, 4, 3]
$10 \sim 15$	[5, 4, 3]	[5, 4, 3]
$15 \sim 25$	[5, 4, 3]	[10, 0, 0]

4. Result and discussion

In this section, numerical simulations are carried out in order to demonstrate the effectiveness of the controllers proposed in this paper. Simulations are divided into two parts, typical trajectory tracking without disturbances including taking-off, hovering and landing and the helix tracking under the external wind guests. Besides, several typical control methods, such as PID, LQR, backstepping (BS) and integral backstepping (integral BS), are applied in the simulations with the purpose of comparing the tracking accuracy and robustness against external disturbances. The 6-DOF dynamic model of the quadrotor helicopter is used in the simulations with the physical parameters shown in Table 1 and the control parameters are designed as follows.

$$\boldsymbol{K} = \begin{bmatrix} \boldsymbol{K}_1 & 0 \\ 0 & \boldsymbol{K}_2 \end{bmatrix}, \quad \boldsymbol{K}_1 = 3\boldsymbol{I}_{3\times3}, \ \boldsymbol{K}_2 = 0.9\boldsymbol{I}_{3\times3},$$

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_1 & 0 \\ 0 & \boldsymbol{A}_2 \end{bmatrix}, \quad \boldsymbol{A}_1 = 1.5\boldsymbol{I}_{3\times3}, \ \boldsymbol{A}_2 = 9\boldsymbol{I}_{3\times3},$$

$$\varepsilon_1 = 10, \qquad \varepsilon_2 = 2,$$

$$\boldsymbol{Q} = \begin{bmatrix} \boldsymbol{Q}_1 & 0 \\ 0 & \boldsymbol{Q}_2 \end{bmatrix}, \quad \boldsymbol{Q}_1 = 5\boldsymbol{I}_{3\times3}, \ \boldsymbol{Q}_2 = 50\boldsymbol{I}_{3\times3}.$$

4.1. Typical trajectory tracking

Considering the characteristics of the quadrotor, the typical trajectory consisted of three parts, i.e., taking-off, hovering and landing. The reference flight trajectory parameters are shown in Table 2.

The performance of trajectory tracking of the proposed controller compared with other control methods is shown in Fig. 4 and Fig. 5. It is obvious that the tracking accuracy of nonlinear control methods (i.e., BS, integral BS and integral BS-SMC) is higher than that of the linear control methods (i.e., PID and LQR) when beginning the maneuvering flight, namely the flight phase of taking-off and landing. The reason is that the nonlinear dynamic characteristics shown in Eqs. (9)–(10) of the quadrotor are fully coupled when accomplishing the movement in two or more directions. Thus, the linear control methods cannot achieve the expected tracking accuracy under the considered input commands on this condition. Since the proposed control strategy (integral BS-SMC) combines the advantages of both backstepping and sliding mode control, its

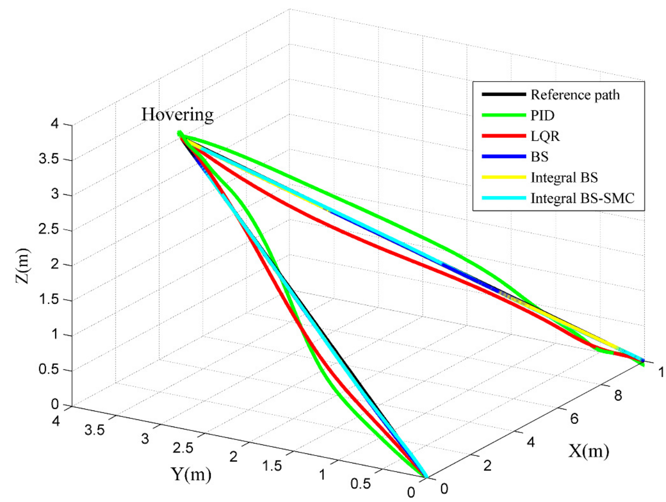

Fig. 4. Trajectory tracking in three dimensional.

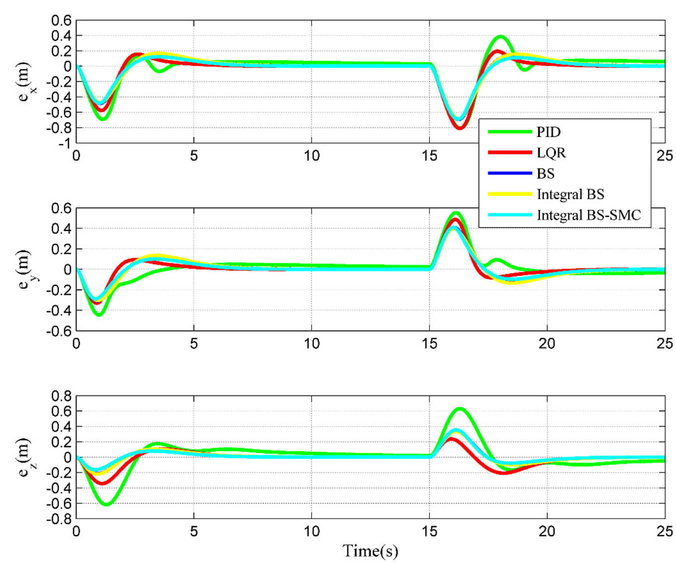

Fig. 5. Position errors in three directions.

position tracking errors shown in Fig. 5 in three directions converge to zero in a shorter time and oscillate in a smaller amplitude than other controllers. On the other hand, during the steady flight, namely the flight phase of hovering in the trajectory, the nonlinear dynamic characteristics are hidden because the attitude and angular velocity approximately equal zero. Thus, all control approaches maintain the same satisfactory level of tracking precision.

It should be pointed out that the improvements of dynamic response are not achieved by the higher control levels but by the form of the control laws. Fig. 6 shows the four practical control inputs of the quadrotor. From the four subfigures and local enlargements at the point of 15 s when the quadrotor begins maneuvering flight, an obvious result can be obtained that the presented control method has a more satisfactory tracking effectiveness than others, even on condition that the amplitude of control inputs is smaller. In order to demonstrate the conclusion more convincingly, we introduce accumulated energy consumption E_M of three control torques characterized by $E_M = \int_{t_0}^{t} \|\boldsymbol{M}^B\| \mathrm{d}\tau$. Since the oscillation times of the control torques of the presented control method are less than those of other methods, the energy consumption shown in Fig. 7 is also satisfactory.

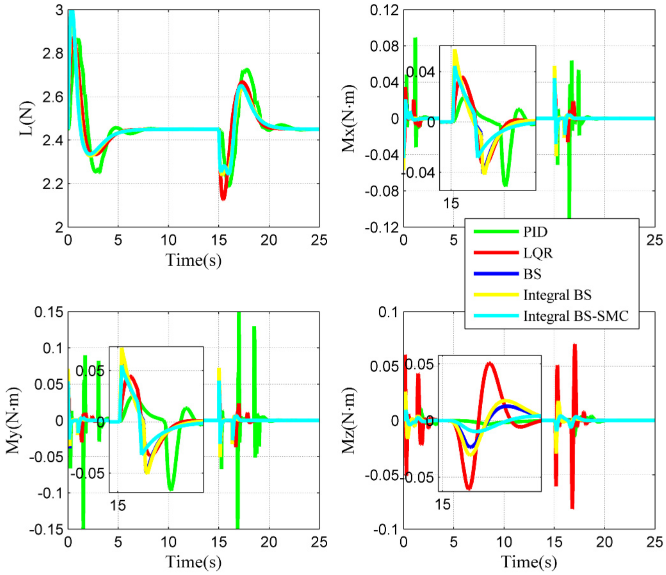

Fig. 6. Control inputs of quadrotor.

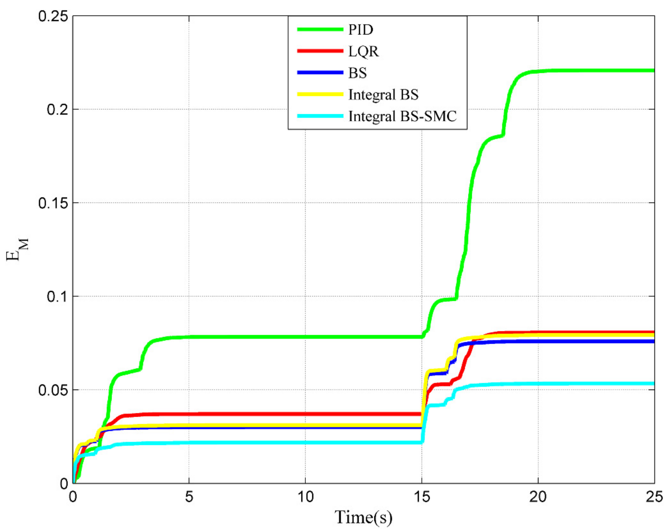

Fig. 7. Energy consumption of five controllers.

4.2. Helix trajectory tracking with disturbances

The reference helix trajectory is defined as follows in the Cartesian space.

$$\begin{cases} x_d = 5\cos(0.1t) + 5 \\ y_d = 5\sin(0.1t) + 5 \\ z_d = 0.3t \end{cases} \tag{48}$$

Assumed that the desired yaw angle equals to 50°, i.e., $\psi_d = 50°$ to illustrate the nonlinear effect caused by it.

The external disturbances referring Eq. (13), including constant slide wind and the stochastic measurement noise corresponding to uniform distribution are modeled as:

$$\begin{cases} \boldsymbol{d}^1 = \begin{bmatrix} \boldsymbol{d}_1^1 \\ \boldsymbol{d}_2^1 \end{bmatrix} = \begin{bmatrix} \boldsymbol{d}_{1,cont}^1 + \boldsymbol{d}_{1,sto}^1 \\ 0 \end{bmatrix}, & 50 \le t \le 60 \text{ s} \\ \boldsymbol{d}^2 = \begin{bmatrix} \boldsymbol{d}_1^2 \\ \boldsymbol{d}_2^2 \end{bmatrix} = \begin{bmatrix} 0 \\ \boldsymbol{d}_{2,cont}^2 + \boldsymbol{d}_{2,sto}^2 \end{bmatrix}, & 150 \le t \le 160 \text{ s} \end{cases} \tag{49}$$

where:

$$\begin{cases} \boldsymbol{d}_{1.cont}^1 = 0.25, \\ -0.01 \le \boldsymbol{d}_{1,sto}^1 \le 0.01, \end{cases} \qquad \begin{cases} \boldsymbol{d}_{2.cont}^2 = 0.01 \\ -0.005 \le \boldsymbol{d}_{2,sto}^2 \le 0.005 \end{cases}$$

And the superscript means the first and second disturbed periods during the flight.

Before performing the numerical simulation, we analyze how the quadrotor system will respond to the constant disturbances. For the first disturbed period, the system responses depend on the quadrotor flight state under the force disturbances. For example, an increasing roll angle and a decreasing pitch angle are required to suit the positive side disturbances if the velocity in horizontal plane of the quadrotor is positive at the same time. During the second disturbed period, the quadrotor can maintain the previous desired attitudes if the tracking precision is high enough, since the moment disturbances can be balanced by supplying corresponding resistant moments. According to the form of the trajectory and the time horizons of these two disturbed periods, the quadrotor flight state and the system responses are summarized in Table 3.

In order to illustrate and compare the performance of the path following under external disturbances, especially the benefits of robustness of the proposed control method, we put another numerical simulation shown in Fig. 8 and the position errors in three directions are illustrated in Fig. 9. It can be seen in local enlargements at 50 s that the trajectory and tracking errors vibrate obviously once disturbances are involved. Compared with the steady-state position errors of the LQR and BS, the position errors of integral BS and integral BS-SMC converge to zero in a finite time. Moreover, the position errors of the proposed controller have smaller overshoots and faster convergence speed. It is worthwhile to note that the position errors during the second disturbed period are much smaller than those under the force disturbances, especially, there is no position error along z-axis due to that the yaw disturbed moment won't influence the altitude according to the quadrotor dynamics. Like the satisfactory performance of the proposed controller in the first disturbed period, its robustness is also excellent under the moment disturbances.

According to the dynamic characteristics of the quadrotor, the robustness of the system not only depends on the translational subsystem, but also on the rotational subsystem, namely the robustness of rotational controller. In order to present the dynamic response of rotational subsystem under external disturbances and to further demonstrate the advantages of the proposed control strategy, we present the attitude commands and the attitude errors in Fig. 10 and Fig. 11. During the first disturbed period, it can be seen in the first subfigure in Fig. 10 that the amplitude of desired roll angle of the proposed control method is the least among these control strategies, otherwise, the linear controllers (PID and LQR) present dramatic waves against the external disturbances. Besides, as with the first subfigure, the desired pitch angle of the integral BS-SMC shown in the second subfigure in

Table 3
System responses against the external disturbances.

	First disturbed period			Second disturbed period		
Flight state		$V_x > 0,\ V_y > 0$			–	
Disturbances	$d_{1x.cont}^1 > 0$	$d_{1y.cont}^1 > 0$	$d_{1z.cont}^1 > 0$	$d_{2x.cont}^2 > 0$	$d_{2y.cont}^2 > 0$	$d_{2z.cont}^2 > 0$
Responses	$\theta\downarrow, \phi\downarrow$	$\theta\downarrow, \phi\uparrow$	$L\downarrow$	$M_x^B\downarrow$	$M_y^B\downarrow$	$M_z^B\downarrow$

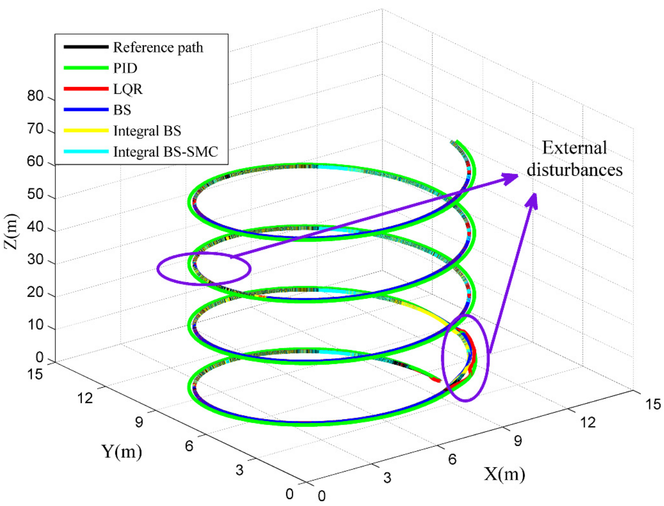

Fig. 8. Helix trajectory tracking under external disturbances.

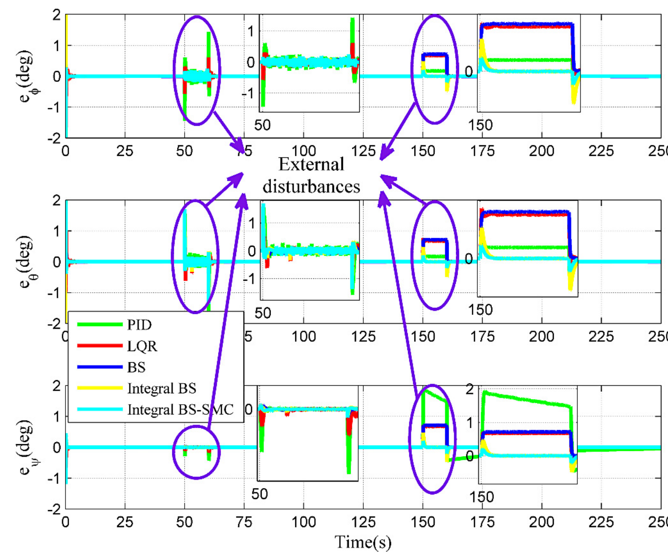

Fig. 11. Attitude errors under external disturbances.

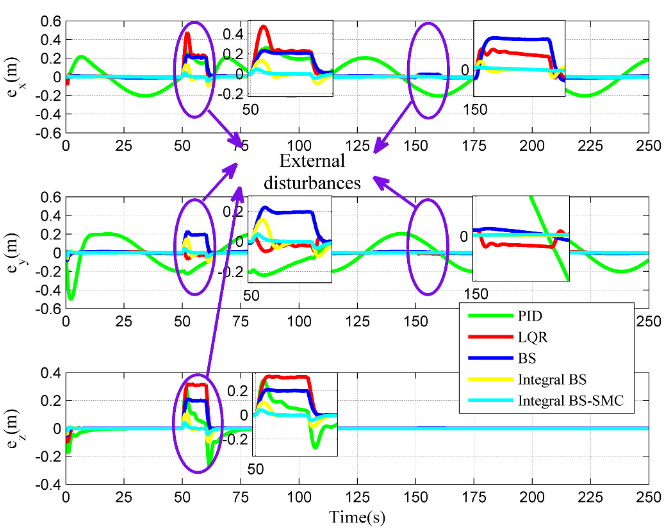

Fig. 9. Position errors under external disturbances.

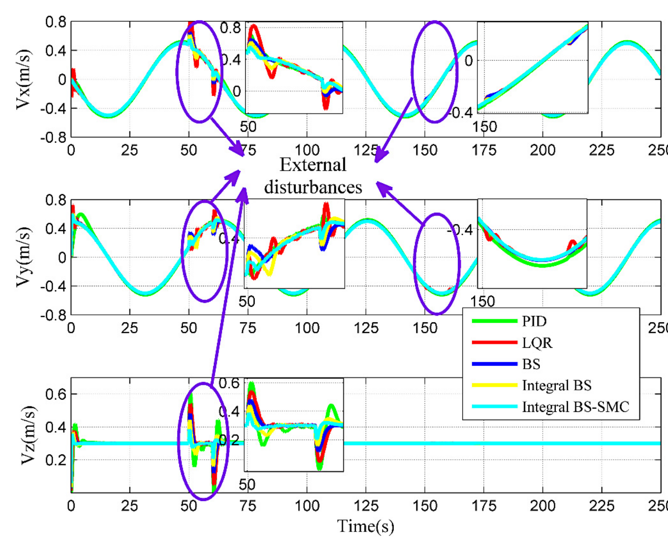

Fig. 12. Velocity under external disturbances.

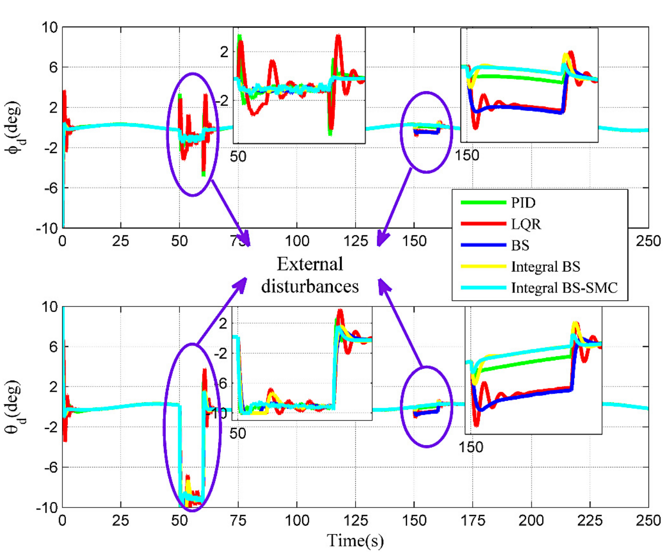

Fig. 10. Attitude commands under external disturbances.

Fig. 10 reaches the steady state fastest with less vibration. During the second disturbed period, the attitude commands of the integral BS and integral BS-SMC return to the commands with no disturbances, which means that just the corresponding resistant moments can reject the disturbances while maintaining the same attitudes on ideal condition. That is the reason why the position errors in the second disturbed period are much smaller than those in the first disturbed period. Moreover, that the command value of pitch angle is larger than that of roll angle is due to the combined effect of the force disturbances along x-axis and y-axis as discussed in Table 3. These simulation results are proved to be consistent with the theoretical analysis shown in Table 3. Besides, in Fig. 11, the angle errors of the integral BS and integral BS-SMC converge to zero while others have steady angle errors, especially under the moment disturbances.

Fig. 12 describes the velocity of the quadrotor in three directions. During the first disturbed period, the velocity of integral BS-SMC converges to a steady state in a shorter time without obvious oscillation, compared with that of other control methods. During the second disturbed period, since the attitude commands almost maintain the value under undisturbed condition, the velocity of the quadrotor is rarely influenced by the disturbances.

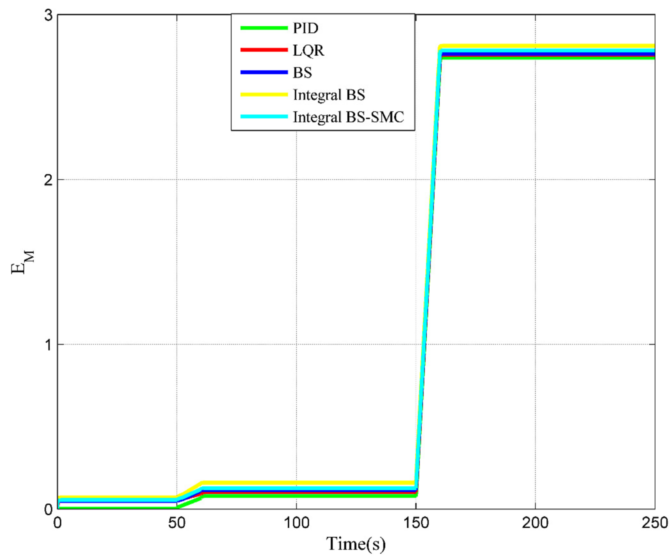

Fig. 13. Energy consumption of five controllers under external disturbances.

At last, it is essential to declare that the excellent robustness of the proposed control strategy is not achieved by a higher control cost but by its superiority derived from the combination of advantages of other control approaches. The accumulated energy consumption E_M of control torques is illustrated in Fig. 13. The differences in control cost among these control methods can be ignored. Hence, the cost of applying the presented controller is the same as others.

5. Conclusions

In this paper, a novel nonlinear control strategy based on integral backstepping combined with sliding mode control is proposed with the goal of tracking the reference path under both the constant and unknown stochastic disturbances. Firstly, the 6-DOF dynamic model of the quadrotor helicopter is set up based on the Newton–Euler formula and the model of disturbances is described subsequently. Next, the controllers are designed for rotational and translational subsystems respectively. Meanwhile, the stabilization ability of the nonlinear controllers are ensured by choosing the proper control inputs. Finally, the numerical simulation results show that the nonlinear control strategy applied in this paper has an obviously better performance than other control approaches no matter in tracking accuracy or robustness against external uncertain disturbances.

Conflict of interest statement

None declared.

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